

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: COMPUTER VISION with OpenCV

COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)*

Assessment Tool Weightage: 20%

Scenario-Based Questions

1. Low Contrast Image

(a) Interpretation

The image has very little difference between the object and the background, making it difficult to distinguish them.

(b) Key Issues

- Poor visibility

- Low contrast between foreground and background
- Loss of important details

(c) Enhancement Technique

Histogram Equalization

(d) Justification

Histogram Equalization spreads pixel intensity values over a wider range, improving contrast and making hidden details visible.

(e) Conclusion

Histogram Equalization improves image quality by increasing contrast and enhancing object visibility.

2. Salt-and-Pepper Noise

(a) Interpretation

The image contains random black and white pixels due to transmission or sensor errors.

(b) Key Issues

- Image appears noisy
- Important details are hidden
- Reduced readability

(c) Filtering Technique

Median Filter

(d) Justification

The Median Filter removes salt-and-pepper noise effectively while preserving image edges better than averaging filters.

(e) Conclusion

Median Filtering restores image quality and improves readability without blurring edges.

3. Object Boundary Detection

(a) Interpretation

The boundaries between different tissues in a medical image need to be identified accurately.

(b) Key Challenges

- Weak edges
- Noise in the image
- Similar intensity values

(c) Edge Detection Method

Canny Edge Detection

(d) Justification

Canny detects edges accurately, reduces noise, and provides thin, continuous boundaries.

(e) Conclusion

Canny Edge Detection is suitable for medical images because it produces precise and reliable boundaries.

4. Gaussian Noise

(a) Interpretation

The image is affected by Gaussian noise, causing random intensity variations.

(b) Key Issues

- Grainy appearance
- Reduced image clarity
- Hidden fine details

(c) Smoothing Filter

Gaussian Filter

(d) Justification

The Gaussian Filter smooths the image and reduces Gaussian noise while preserving overall image structure.

(e) Conclusion

Gaussian Filtering improves image quality by reducing random noise and enhancing clarity.

5. Threshold Selection Problem

(a) Interpretation

The image requires segmentation, but selecting the correct threshold manually is difficult.

(b) Key Issues

- Different lighting conditions
- Overlapping intensity values
- Poor segmentation accuracy

(c) Thresholding Method

Otsu's Thresholding

(d) Justification

Otsu's method automatically finds the optimal threshold by minimizing intra-class variance, making segmentation more accurate.

(e) Conclusion

Otsu's Thresholding provides automatic and effective image segmentation without manual threshold selection.

These five questions are among the easiest and most commonly asked topics in image enhancement and segmentation, making them good choices for your assessment.

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues

Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand